

# Chemin Ahn

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Birth: 1998.12.22

Portfolio: <https://chemx3937.github.io/>



## Education

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- Bachelor of Science in Mechanical Engineering  
Chungang University, Seoul, South Korea
  - Major in Mechanical Engineering
  - 3.84 / 4.50 GPA (CUM LAUDE)
- Master Course of Science in Mechanical Engineering  
Sungkyunkwan University, Suwon, South Korea
  - Major in Mechanical Engineering
  - Robotics Innovatory Lab (Prof. Hyouck Ryeol Choi)

## Research Interest

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- Robotics
- Imitation Learning
- Teleoperation
- Compliance Control
- VLA
- Reinforcement Learning
- Computer Vision

## Projects

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### Master Course

- **Teleoperation:**
  - Single Arm Leader Arm  
w/ Teleoperation + Force Feedback + Gripper Feedback + Gravity Compensation  
(<https://youtu.be/dhNTIGTX53Q?si=il2BSdpKIuVks4Q8>)
  - Dual Arm Leader arm  
w/ Teleoperation + Force Feedback + Gravity Compensation  
(<https://youtu.be/Q0x2Fh5BNF8?si=XVNefWEzQ33wNWMp>)
  - RB Y-1 Master Arm Force Feedback Teleoperation  
([https://youtu.be/adola9Bo\\_tA?si=rqvpB0QddU2BJhsM](https://youtu.be/adola9Bo_tA?si=rqvpB0QddU2BJhsM))
  - Using VR Tracker  
(<https://youtu.be/ToG8sNmdli0?si=a3Xqx6uPEkklisuQ>)
  - Using Visual SLAM based VR Tracker  
(<https://youtu.be/yQAdfKhJR1M?si=dOepBiyha9SJi26N>)

- **Imitation Learning**

- Dual Arms + Hands  
(<https://youtu.be/Vs-ICyzwaLc>)
- Dual Arms  
(<https://youtu.be/WTx4ySIACAU>)
- One Arm  
([https://youtu.be/VPG-\\_bH2SSY](https://youtu.be/VPG-_bH2SSY))
- Multimodal (Vision + Force) Imitation Learning – On going

## **Intern & Others**

- Doosan Robotics Robot Control Team intern: 2024.07.01~10.04
  - Project: Visual Servoing Package  
([https://youtu.be/9UFAdaa\\_PR8?si=LtkzL9kk2eVBcLp8](https://youtu.be/9UFAdaa_PR8?si=LtkzL9kk2eVBcLp8))
- AI-Robotics Academy (KG-Kairos): 2023.12.19~24.06.17
  - Project: assisting pharmaceutical preparation and autonomous delivery using cobot and AMR  
([https://youtu.be/-mTXpCWZacI?si=YXC8AwRQ8FV\\_uN9S](https://youtu.be/-mTXpCWZacI?si=YXC8AwRQ8FV_uN9S))

## **Bachelor's degree**

- Finding the optimal sensor for AMR and analyzing driving characteristics by algorithm combinations  
(<https://youtu.be/whuUySDWIwA?si=vPVX8IQul7cd9p3q>)
- Undergraduate research student: 2022.12.22~2023.02.28
  - Design light weight passive wearable suit for logistic workers
- Design light weight battle robot
- Maze escape using OD based autonomous drone

## **Awards**

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- KRoC 2026 RED Show @ KROS (KRoC 2026) – 1<sup>st</sup> prize (2026.02)
- Capstone Design Contest @ Chungang University – 2<sup>nd</sup> prize (2024.06)
- AI-Robotics Academy @ KG-Kairos (KG ICT) – 2<sup>nd</sup> prize (2024.06)
- Cum Laude @ Chungang University (2025.02)

## **Patent**

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- Force Feedback Leader arm Teleoperation – Patent Pending

## **Imitation Learning Policies**

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- ACT
- Diffusion Policy
- DiT-Policy
- Adaptive Compliance Policy (ACP)

## Tool Skills

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- Language & Framework: Python, ROS 1, 2, MATLAB
- Libraries: Pytorch, TensorFlow, OpenCV, Numpy, Pandas, matplotlib
- Simulation Tools: Isaac Sim, Isaac Lab, Gazebo, Ignition Gazebo
- Manipulator: Doosan Robotics, Rainbow Robotics
- Other: Git, SolidWorks

## Language Skills

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- English: fluent (reading, listening); intermediate (speaking, writing)
  - TOEIC Speaking: 150(IH)

## Additional Link

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- Portfolio: <https://chemx3937.github.io/>
- Github: <https://github.com/Chemx3937>
- Youtube: <https://www.youtube.com/@%EC%B3%84-r3i>
- LinkedIn: <https://www.linkedin.com/in/chem1222/>

## Related Courses

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- Intelligent Robotics
- Robot Reinforcement Learning
- Modern Control Systems
- Human-Robot Collaboration
- Dynamics
- System Dynamics
- Control Systems Engineering
- Mechatronics
- Robotics
- Numerical Methods
- Design and Analysis of Experiments
- System Control Experiments
- Visual Programming